

Shay Mordechai

Security Researcher | Systems, Network Internals & Embedded Security

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Summary

Security Researcher specializing in **Application Security**, **Cloud Architecture**, and **Protocol Analysis**. Proven experience in performing **Security Design Reviews** and **Threat Modeling** for enterprise-grade environments. Skilled in uncovering vulnerabilities through manual research and SAST/SCA tools, focusing on hardening API endpoints and developing custom mitigation strategies.

Professional Experience

Independent Security Researcher & Architect | *MyLeads AI* | Jan 2026 – Present

- **Zero-Trust Infrastructure:** Engineered a "Dark Server" AWS architecture (0 inbound ports), routing traffic via Cloudflare Tunnels and Rootless Podman.
- **Application Security Auditing:** Audited B2B architectures to identify and mitigate web-based vulnerabilities and secure API endpoints against advanced exploitation techniques.
- **Wasm DLP Firewall:** Engineered a Rust WebAssembly filter for Envoy Proxy to redact sensitive egress traffic, preventing Information Disclosure.

Application Security Intern | *Israel Tax Authority* | Aug 2024 – Jul 2025

- **Defense-in-Depth:** Partnered with R&D teams to perform Security Design Reviews and Threat Modeling. Hardened IBM DataPower gateways via strict JSON schemas.
- **AppSec & Remediation:** Integrated Checkmarx SAST/SCA into 15+ CI/CD pipelines. Triaged findings and trained teams to mitigate vulnerabilities.

IDF Combat Soldier (Hesder) | *Combat Intelligence Collection Corps* | Aug 2016 – Apr 2018 - Executed covert reconnaissance and visual anomaly detection for actionable operational target profiling.

Education

B.Sc. Computer Science | *Jerusalem College of Technology* | 2021-2025 | GPA: 89

- **Coursework:** Network Analysis (97), InfoSec (90), Advanced Malware Analysis (IoCs, Yara - In Progress).
- **Reverse Engineering (93):** Conducted static/dynamic analysis of x86/x64 binaries and mobile components via **IDA/x32dbg** to reverse-engineer logic and bypass anti-debugging.

Technical Projects & Research

- **Hardware Guard (Rust/ESP32):** Developing a bare-metal gateway implementing DPI to mitigate L7 attacks (DNS Rebinding) based on previous exploit research validated by TP-Link & Palo Alto PSIRT.
- **SBCL Compiler Vuln Research:** Developed custom scripts and tools to automate vulnerability discovery within third-party dependencies, reporting 3 Zero-Days.
- **V8 Engine Internals:** Published analysis on V8 JIT vulnerabilities, utilizing detours and runtime hooking to bypass CFI via RWX memory.

Technical Skills

- **Security & Research:** Reverse Engineering, Exploit Development, Fuzzing, Runtime Code Manipulation, Bug Bounty.
- **Languages & Low-Level:** Python, C/C++, x86 Assembly, Bash, Rust, OS Internals.